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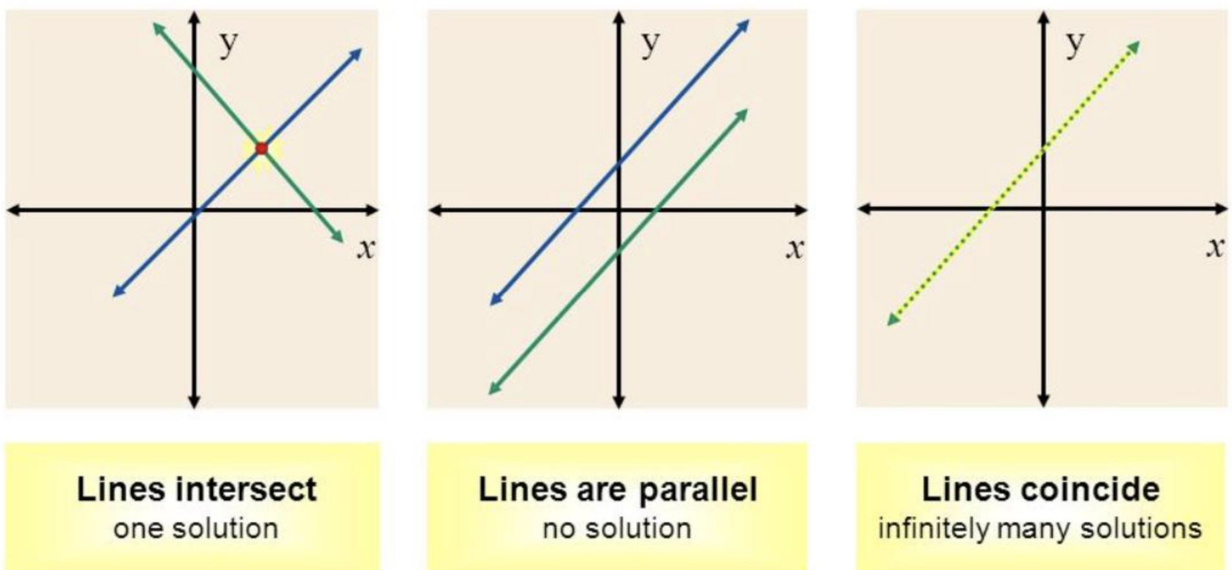
System of Linear Equation

Relation between two straight lines:



Interpretation:

❖ 2 or more straight lines constitute a system.

Relation between two lines,

- ➢ If two straight lines intersect each other, then create 1 common point of them.
- ➢ If two straight lines are parallel, then there is no common points.
- ➢ If one straight lines overlap to another straight lines, then there is infinity common points.

Some necessary concept:

- Every equation like $ax+by+c=0$ represents a straight line.
- The equation $ax+by+c=0$ is a linear equation in 2 variables.
- More than one equations will make a system of linear equations.
  The equations of the system may have
  1. Only 1 common solution or unique solution.
  2. No common solution or no solutions.
  3. Infinite solutions or many solutions.

GENERALIZATION:

- The equation
$$a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_nx_n = 0 \dots \dots \dots (1)$$
is a linear equation in n variables.
- A SLE consists of m linear equations, each in n variables. It may be written as follows.
$$\left.\begin{array}{l} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2 \\ \dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots \\ \dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n = b_m \end{array}\right\} \dots \dots \dots (2)$$
- The SLE will be called **homogeneous** if the RHS of each equations are equal to zero.
- The SLE will be called **non-homogeneous** if the RHS of each equations are not equal to zero.

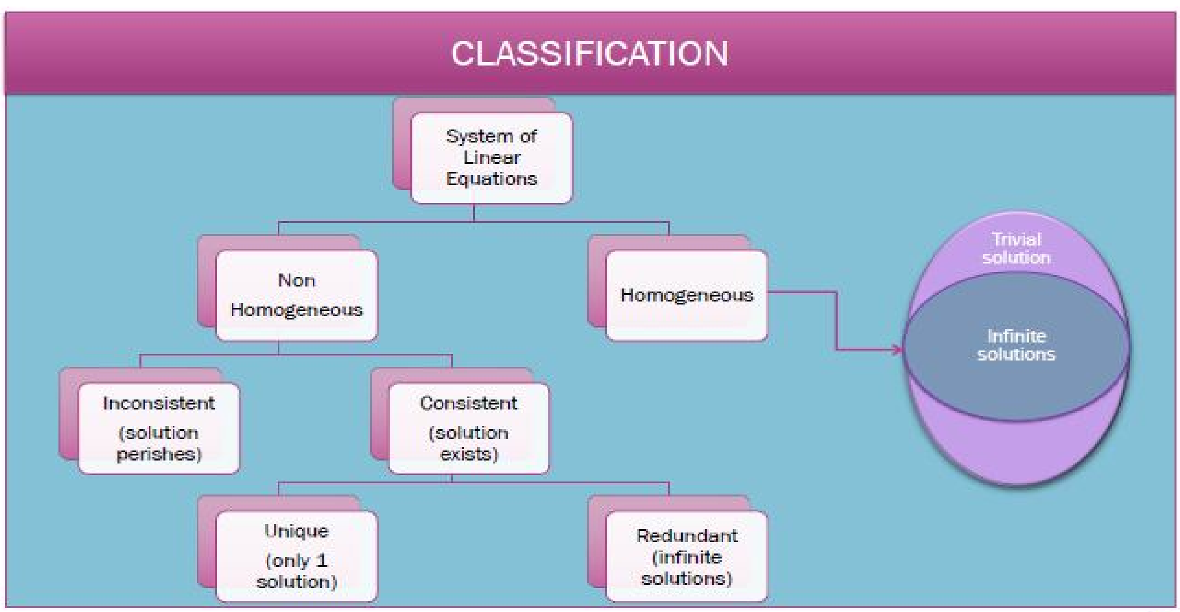
MATRIX FORM OF SLE:

Matrix Form

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

$$AX = b$$

Where A: coefficient matrix; X: Unknown or Solution Matrix; b: Constant Matrix.



METHODS FOR SOLUTION:

- Matrix Method
- Determinant Method
- REM/RREM Method

Example: Solve the System of Linear Equations,

$$-5a+3b-3c=17$$
$$4a-3b+8c=9$$
$$8a+5b=-2$$

Solution: The given system is converted to matrix form,

$$\begin{bmatrix} -5 & 3 & -3 \\ 4 & -3 & 8 \\ 8 & 5 & 0 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix}=\begin{bmatrix} 17 \\ 9 \\ -2 \end{bmatrix}$$

$$\Rightarrow AX=b \quad \text{[Say]}$$

$$\Rightarrow A^{-1}AX=A^{-1}b$$

$$\Rightarrow IX=A^{-1}b$$

$$\Rightarrow X=A^{-1}b$$

$$\Rightarrow \begin{bmatrix} a \\ b \\ c \end{bmatrix}=\frac{1}{260}\begin{bmatrix} -40 & -15 & 15 \\ 64 & 24 & 28 \\ 44 & 49 & 3 \end{bmatrix}\begin{bmatrix} 17 \\ 9 \\ -2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a \\ b \\ c \end{bmatrix}=\begin{bmatrix} -13/4 \\ 24/5 \\ 91/20 \end{bmatrix}$$

$$\therefore a=-\frac{13}{4};\ b=\frac{24}{5};\ c=\frac{91}{20}$$

Limitations:

1. If M is not square, this method can’t be applied.
2. Does not work for the system when |M|= 0.
3. Becomes too much lengthy when M is 4x4, 5x5, or so on.

WHY REM/RREM METHOD?

All problems related to the solution are overcome when we apply the general REM/RREM method, which depends on the REF/RREF method, to solve a SLE.

SOLUTION OF NON-HOMOGENEOUS SLE

The given system of linear equations :

$$\begin{aligned} 3x_1 + 6x_2 + x_3 &= 16 \\ 2x_1 + 4x_2 + 3x_3 &= 13 \\ x_1 + 3x_2 + 2x_3 &= 16 \end{aligned}$$

Augmented Matrix

$$(A \mid b) = \left(\begin{array}{ccc|c} 3 & 6 & 1 & 16 \\ 2 & 4 & 3 & 13 \\ 1 & 3 & 2 & 16 \end{array}\right)$$

REF

$$\left(\begin{array}{ccc|c} 3 & 6 & 1 & -1 \\ 0 & 3 & 5 & 11 \\ 0 & 0 & 7 & 7 \end{array}\right)$$

Rank(Augmented Matrix) = 3
Rank(Coefficient Matrix) = 3
No. of Variables = 3

All Equal

Consistent System!!!
Unique Solution!!!

REF

$$\left(\begin{array}{ccc|c} 3 & 6 & 1 & -1 \\ 0 & 3 & 5 & 11 \\ 0 & 0 & 0 & 7 \end{array}\right)$$

Rank(Augmented Matrix) = 3
Rank(Coefficient Matrix) = 2

Not Equal

No. of Variables = 3

Inconsistent System!!!
No Solution!!!

REF

$$\left(\begin{array}{ccc|c} 3 & 6 & 1 & -1 \\ 0 & 3 & 5 & 11 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Rank(Augmented Matrix) = 2
Rank(Coefficient Matrix) = 2

Both Equal

No. of Variables = 3

Consistent System!!!
Many Solution!!!

CONSISTENT SLE: **UNIQUE SOLUTION**

Solve the system of linear equations :

$$\begin{aligned} 3x_1 + 6x_2 + x_3 &= 16 \\ 2x_1 + 4x_2 + 3x_3 &= 13 \\ x_1 + 3x_2 + 2x_3 &= 16 \end{aligned}$$

Solution: The augmented matrix of the system is:

$$(A \mid b) = \left(\begin{array}{ccc|c} 3 & 6 & 1 & 16 \\ 2 & 4 & 3 & 13 \\ 1 & 3 & 2 & 16 \end{array}\right) \sim \left(\begin{array}{ccc|c} 3 & 6 & 1 & 16 \\ 0 & 0 & 7 & 7 \\ 0 & 3 & 5 & 32 \end{array}\right) \quad \begin{array}{l} R_2 \longrightarrow 3R_2 - 2R_1 \\ R_3 \longrightarrow 3R_3 - R_1 \end{array}$$

$$\sim \left(\begin{array}{ccc|c} 3 & 6 & 1 & 16 \\ 0 & 3 & 5 & 32 \\ 0 & 0 & 7 & 7 \end{array}\right) \quad R_2 \leftrightarrow R_3$$

REF

Rank(Augmented Matrix) = 3
Rank(Coefficient Matrix) = 3
No. of Variables = 3

All Equal

Consistent System!!!
Unique Solution!!!

No. of Free Variables = No. of Unknown (Variables) – No. of Pivot = 3 – 3 = 0

Hence the system is consistent and therefore the system has unique solution.

By back substitution:

$$7x_3 = 7 \implies x_3 = 7/7 = 1$$

$$3x_2 + 5x_3 = 32 \implies x_2 = \frac{1}{3}(32 - 5x_3) = \frac{1}{3}(32 - 5 \times 1) = 9$$

$$3x_1 + 6x_2 + x_3 = 16 \implies x_1 = \frac{1}{3}(16 - 6x_2 - x_3) = \frac{1}{3}(16 - 6 \times 9 - 1) = -13$$

CONSISTENT SLE: **MANY SOLUTION**

Solve the system of linear equations:

$$2x - 4y + 6z = 8$$
$$3x + 2y + z = 4$$
$$-x + 2y - 3z = -4$$

Rank(Augmented Matrix) = 2
Rank(Coefficient Matrix) = 2
Both Equal
No. of Variables = 3

Solution: The augmented matrix of the system is:

$$(A \mid b) = \left(\begin{array}{ccc|c} 2 & -4 & 6 & 8 \\ 3 & 2 & 1 & 4 \\ -1 & 2 & -3 & -4 \end{array}\right) \sim \underbrace{\left(\begin{array}{ccc|c} 2 & -4 & 6 & 8 \\ 0 & 16 & -16 & -16 \\ 0 & 0 & 0 & 0 \end{array}\right)}_{\text{REF}} \begin{array}{l} R_2 \longrightarrow 2R_2 - 3R_1 \\ R_3 \longrightarrow 2R_3 + R_1 \end{array}$$

Consistent System!!!
Many Solution!!!

No. of Free Variables = No. of Unknown (Variables) – No. of Pivot = 3 – 2 = 1

Hence the system is consistent and therefore the system has many solution (redundant solution).

Here the free variable is z and let $z = t$ $(where\ t\ is\ arbitrary\ constant)$

By back substitution: **(General Solution)**

$$z = t$$
$$0.x + 16y - 16z = -16 \Rightarrow y = \frac{1}{16}(-16 + 16z) = (-1 + z) = (-1 + t) = t - 1$$
$$2x - 4y + 6z = 8 \Rightarrow x = \frac{1}{2}(4y - 6z) = \frac{1}{2}[8 + 4(t - 1) - 6t] = \frac{1}{2}(4 - 2t) = 2 - t$$

CONSISTENT SLE: **MANY SOLUTION (CONT…)**

Solve the system of linear equations:

$$2x - 4y + 6z = 8$$
$$3x + 2y + z = 4$$
$$-x + 2y - 3z = -4$$

Solution(continue):

By back substitution: **(General Solution)**

$$z = t$$
$$0.x + 16y - 16z = -16 \Rightarrow y = \frac{1}{16}(-16 + 16z) = (-1 + z) = (-1 + t) = t - 1$$
$$2x - 4y + 6z = 8 \Rightarrow x = \frac{1}{2}[8 + 4(t - 1) - 6t] = \frac{1}{2}(4 - 2t) = 2 - t$$

General Solutions are:

$$\boldsymbol{x = 2 - t, \qquad y = t - 1, \qquad z = t}$$

Particular Solution:

For particular solution let $t = 1$ then

$$x = 1, \qquad y = 0, \qquad z = 1$$

INCONSISTENT SLE: NO SOLUTION

Solve the system of linear equations by using gaussian elimination method:

$$\begin{aligned} x_1 + 2x_2 - 3x_3 &= -1 \\ 5x_1 + 3x_2 - 4x_3 &= 2 \\ 3x_1 - x_2 + 2x_3 &= 7 \end{aligned}$$

Solution: The augmented matrix of the system is:

$$(A \mid b) = \left(\begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 5 & 3 & -4 & 2 \\ 3 & -1 & 2 & 7 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 0 & -7 & 11 & 7 \\ 0 & -7 & 11 & 10 \end{array}\right) \quad \begin{array}{l} R_2 \to R_2 - 5R_1 \\ R_3 \to R_3 - 3R_1 \end{array}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 0 & -7 & 11 & 7 \\ 0 & 0 & 0 & 3 \end{array}\right) \quad R_3 \to R_3 - R_2$$

$\boxed{\text{REF}}$

Rank(Augmented Matrix) = 3

Rank(Coefficient Matrix) = 2

Not Equal

No. of Variables = 3

Inconsistent System!!!

No Solution!!!

The system contains an equation of the form $0 = 3$ (which is impossible) .

Hence the system is inconsistent and therefore the system has no solution.

Exercise

- When does an SLE become consistent?
- How many free variables do exist for a unique solution of SLE?
- An SLE can't have all free variables. True or false?
- Write down the condition of consistency of an SLE?
- Write down the condition of an SLE has a unique solution and when it is redundant.

Solve the following SLE.

1. $$\begin{aligned} 3x - 4y + 5z &= 0 \\ x + y - 7z &= 0 \\ 6x + y - z &= -1 \end{aligned}$$

2. $$\begin{aligned} x + y + z &= 0 \\ -x + y - z &= 9 \\ 4x - 7y + 9z &= 1 \end{aligned}$$

3. $$\begin{aligned} 7x - y + z + t &= 0 \\ x + 9y - z &= -8 \\ 2x + 5z - 3t &= 4 \\ x - y - z - t &= -5 \end{aligned}$$

4. $$\begin{aligned} a + b - c - d &= 7 \\ -2a + b - d &= 0 \\ a - b + c - d &= 9 \\ 5a + 4b + 3c + 2d &= 1 \end{aligned}$$